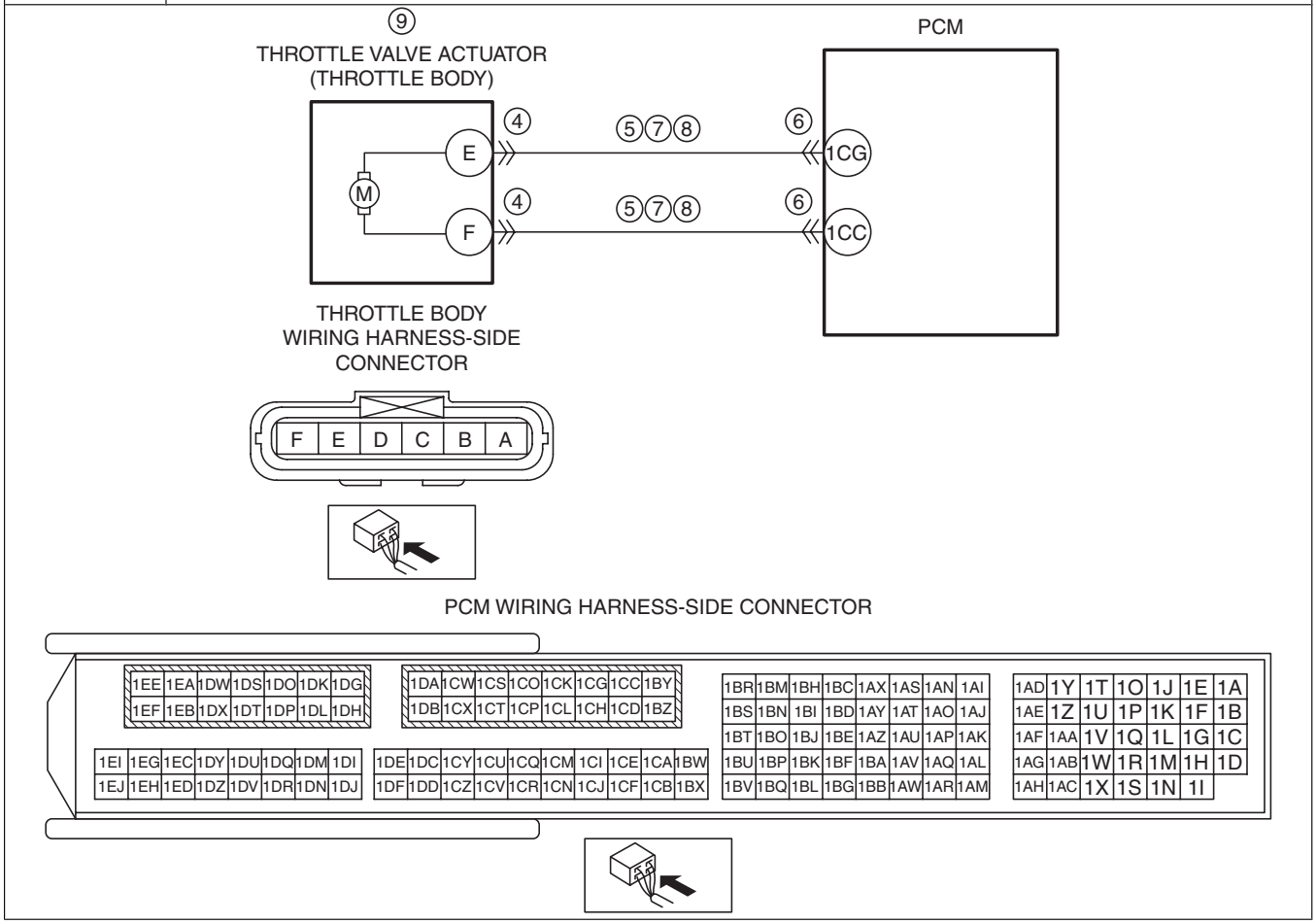
DTC P0638:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0638:00	Throttle valve actuator control range/performance problem
DETECTION CONDITION	• The PCM compares the actual TP with the target TP. If the difference is more than the threshold value, the PCM determines that there is a throttle valve actuator control range/performance problem. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Restricts the upper limit of the engine speed. • Stops the drive-by-wire control (throttle valve is open at **approx. 8 °** by return spring force).
POSSIBLE CAUSE	• Throttle body connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Throttle body terminal E—PCM terminal 1CG — Throttle body terminal F—PCM terminal 1CC • PCM connector or terminals malfunction • Short to power supply in wiring harness between the following terminals: — Throttle body terminal E—PCM terminal 1CG — Throttle body terminal F—PCM terminal 1CC • Open circuit in wiring harness between the following terminals: — Throttle body terminal E—PCM terminal 1CG — Throttle body terminal F—PCM terminal 1CC • Throttle valve actuator malfunction • Throttle valve malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Perform the Freeze Frame PID Data Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the DTC P0638:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
3	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
4	**INSPECT THROTTLE BODY CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the throttle body connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
5	**INSPECT THROTTLE VALVE ACTUATOR CIRCUIT FOR SHORT TO GROUND** • Verify that the throttle body connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Throttle body terminal E — Throttle body terminal F • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Throttle body terminal E—PCM terminal 1CG — Throttle body terminal F—PCM terminal 1CC **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to Step 10.
		No	Go to the next step.
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.

STEP	INSPECTION		ACTION
7	**INSPECT THROTTLE VALVE ACTUATOR CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the throttle body and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the following terminals (wiring harness-side): — Throttle body terminal E — Throttle body terminal F • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Throttle body terminal E—PCM terminal 1CG • Throttle body terminal F—PCM terminal 1CC **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 10.
8	**INSPECT THROTTLE VALVE ACTUATOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the throttle body and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side): — Throttle body terminal E—PCM terminal 1CG — Throttle body terminal F—PCM terminal 1CC • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Throttle body terminal E—PCM terminal 1CG • Throttle body terminal F—PCM terminal 1CC **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
9	**INSPECT THROTTLE VALVE** • Perform the Electronic Control Throttle Operation Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the throttle body, then go to the next step. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.